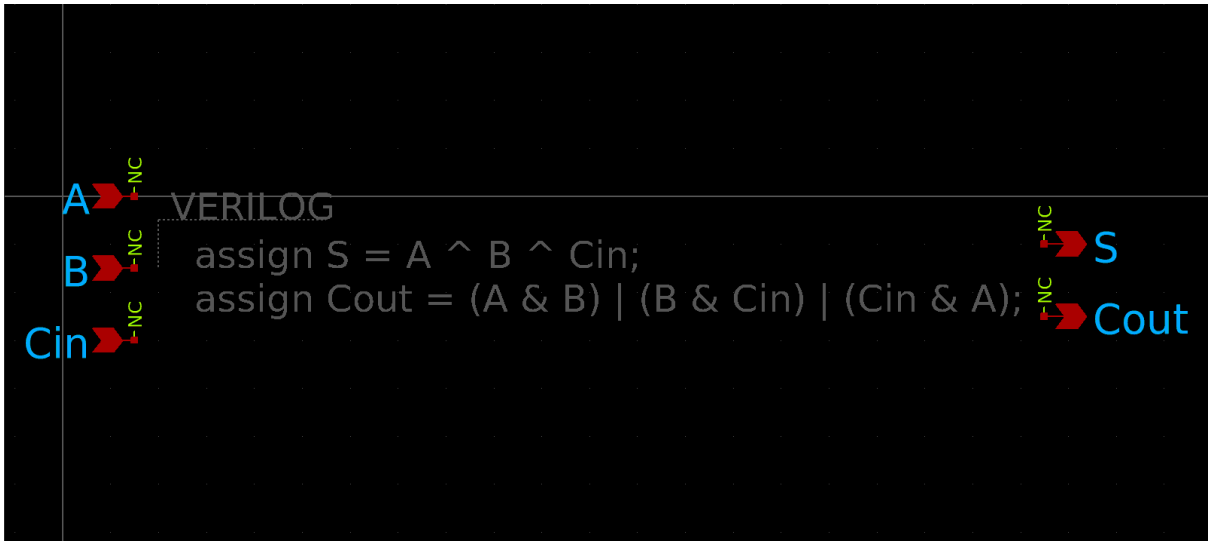
EE5311: Digital IC Design
Tutorial 9

Putta Shravya
Roll Number: EE22B032

October 16, 2025

Question 1

Schematic

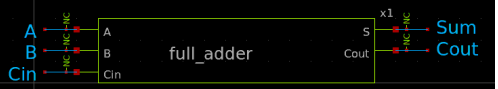


Testbench

```
VERILOG

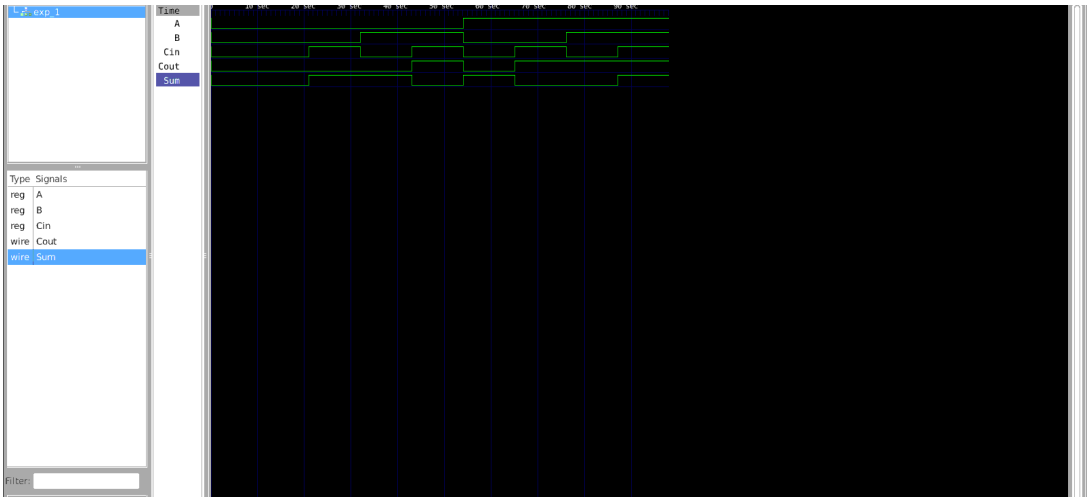
  initial begin
   $dumpfile("dumpfile.vcd");
   $dumpvars(0,A,B,Cin,Sum,Cout);
   A=0;
   B=0;
   Cin=0;
  end

  integer x;
  always begin
  for(x=0;x<8;x=x+1)begin
   #10
   {A,B,Cin} = x;
   #1
   assert ({Cout,Sum} == (A+B+Cin))
     else $error("Failed: %0d+%0d+%0d != {%0d,%0d}",A,B,Cin,Sum,Cout);
   end
   #10
   $finish;
  end
```



Testbench

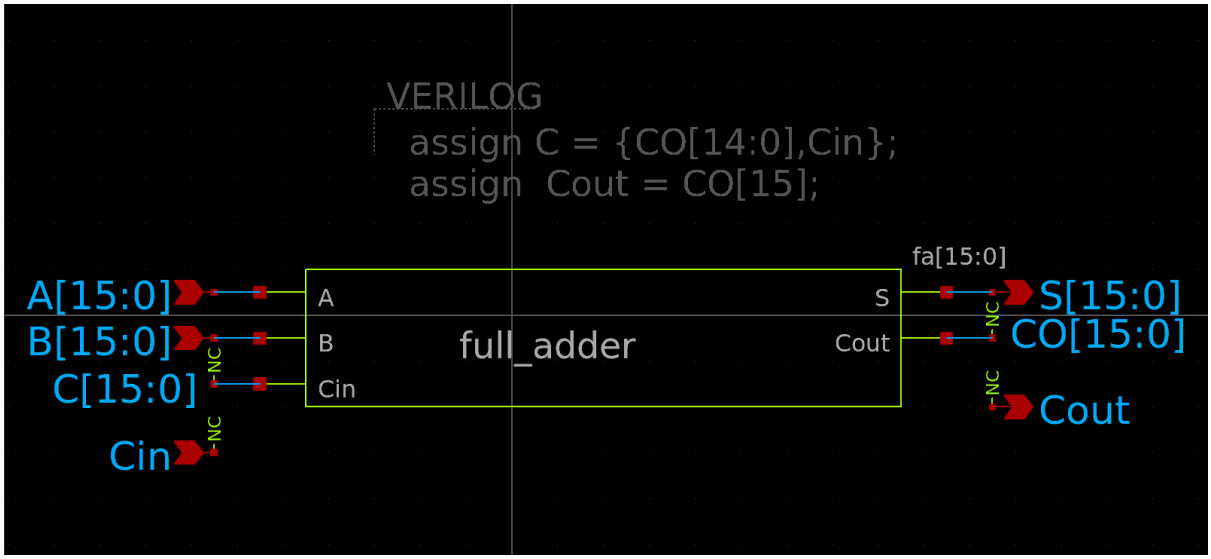
GTKWaves



GTKWaves

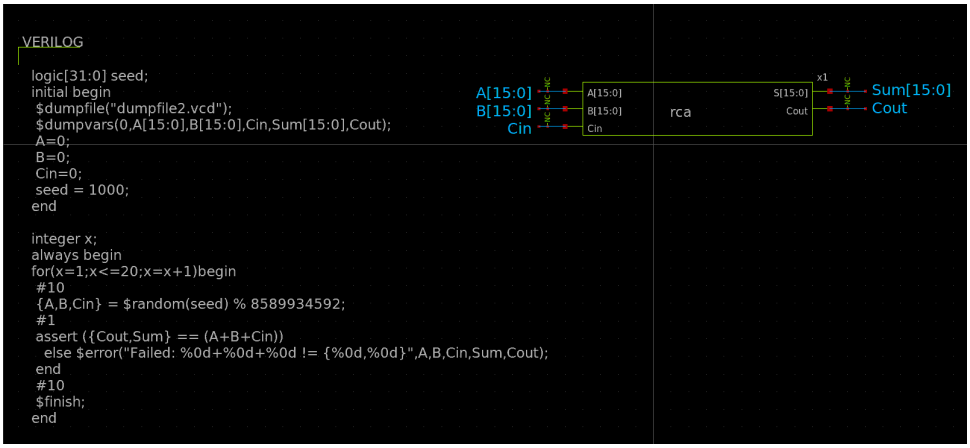
Question 2

Schematic



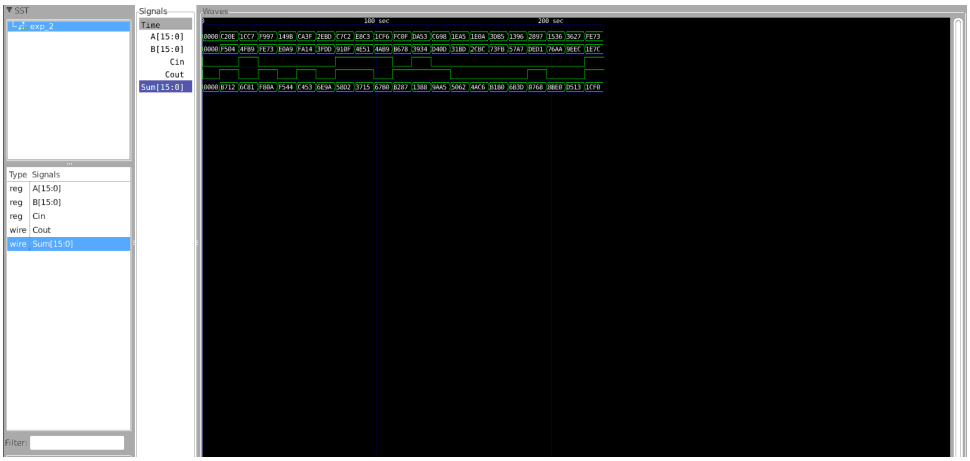
Schematic

Testbench



Testbench

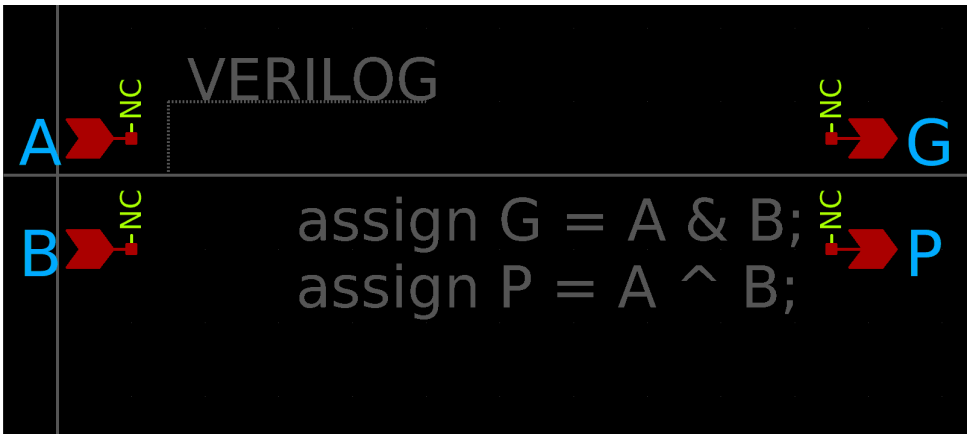
GTKWaves



GTKWaves

Question 3

Propagate/Generate



Prop/Gen

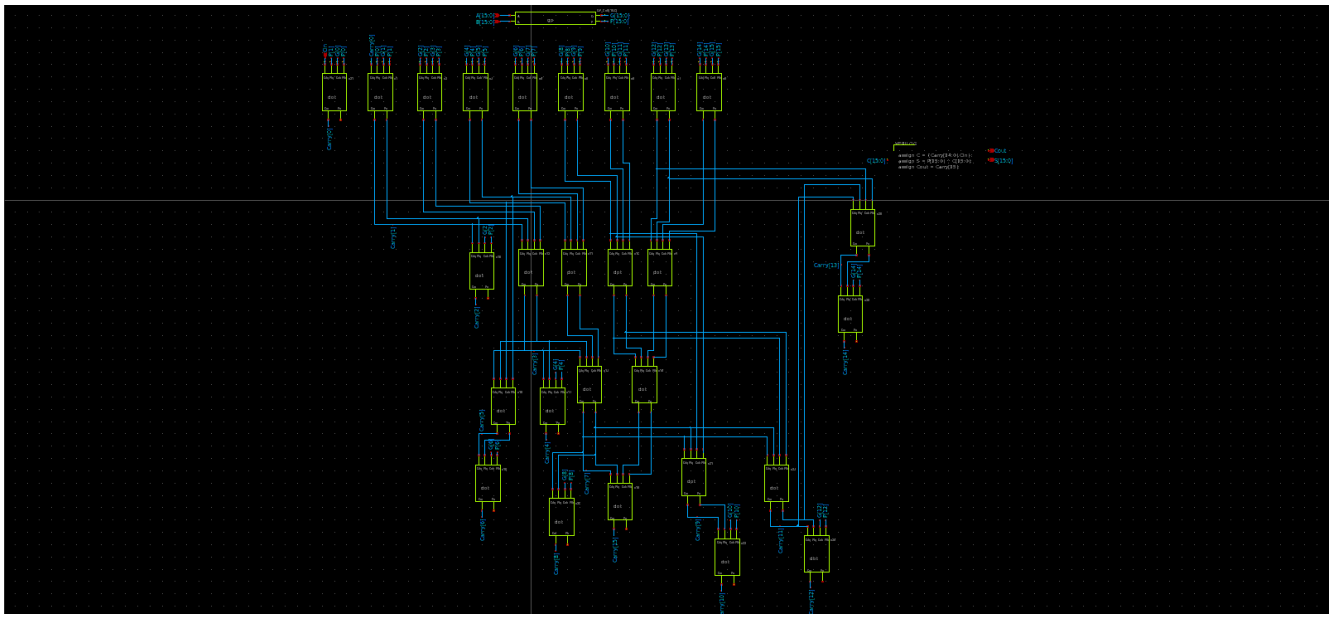
Dot



Dot

Part A

Schematic



Schematic

Testbench

```
VERILOG

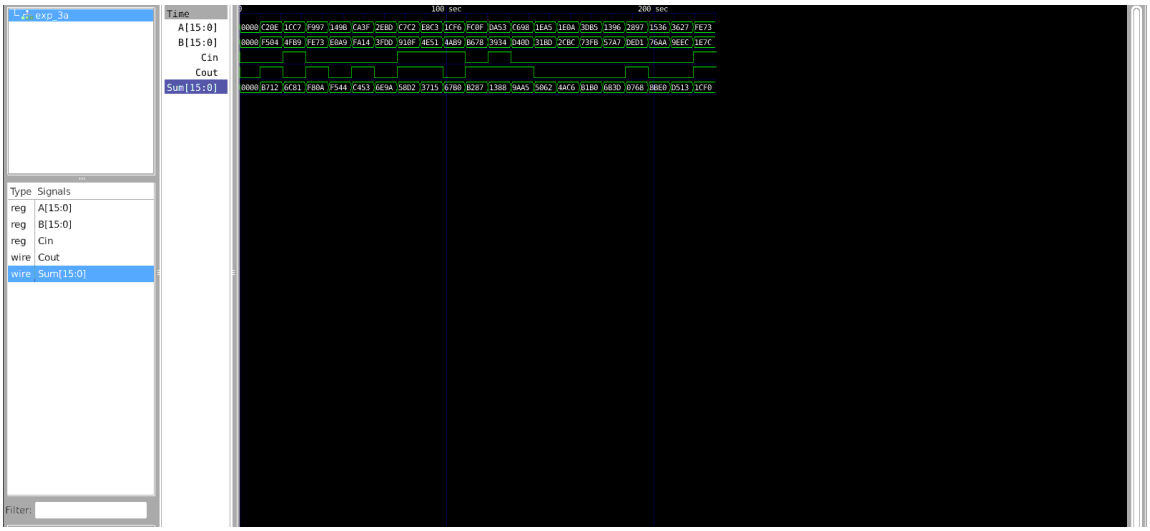
logic[31:0] seed;
initial begin
 $dumpfile("dumpfile3.vcd");
 $dumpvars(0,A[15:0],B[15:0],Cin,Sum[15:0],Cout);
 A=0;
 B=0;
 Cin=0;
 seed = 1000;
end

integer x;
always begin
for(x=1;x<=20;x=x+1)begin
 #10
 {A,B,Cin} = $random(seed) % 8589934592;
 #1
 assert ({Cout,Sum} == (A+B+Cin))
   else $error("Failed: %0d+%0d+%0d != {%0d,%0d}",A,B,Cin,Sum,Cout);
 end
 #10
 $finish;
end
```

A[15:0]
B[15:0]
Cin
A[15:0]
B[15:0]
Cin
bruntkung
Cout
S[15:0]
x1
Cout
Sum[15:0]

Testbench

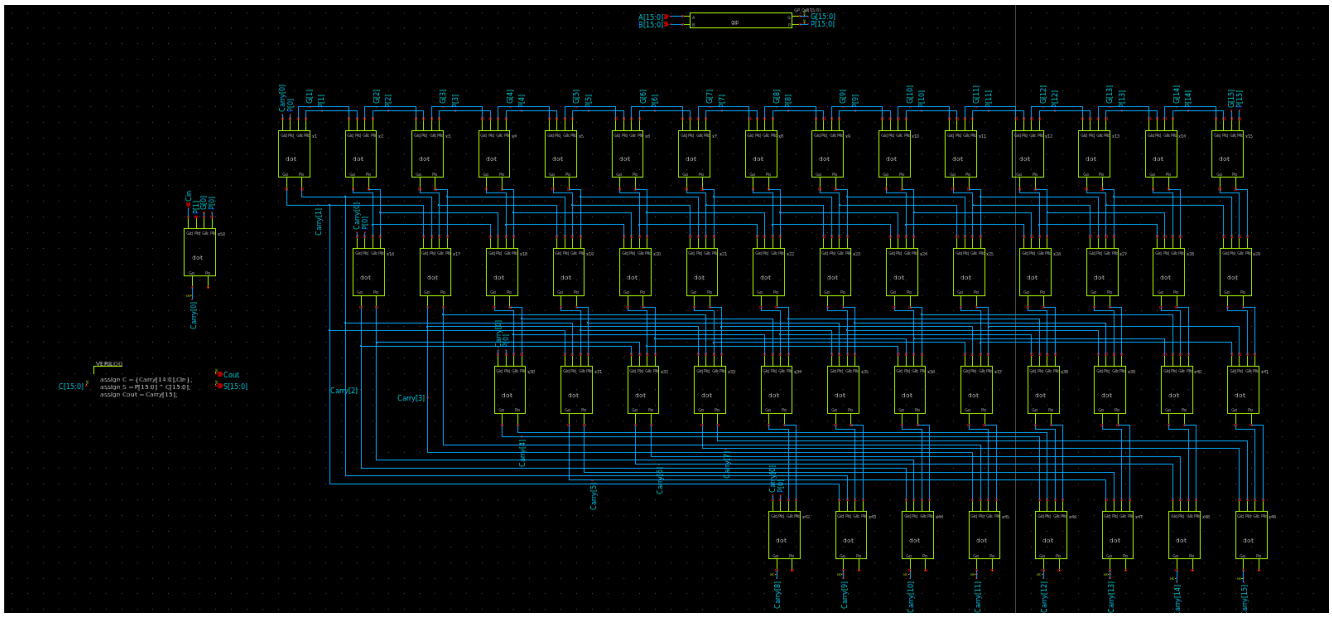
GTKWaves



GTKWaves

Part B

Schematic



Schematic

Testbench

```
VERILOG

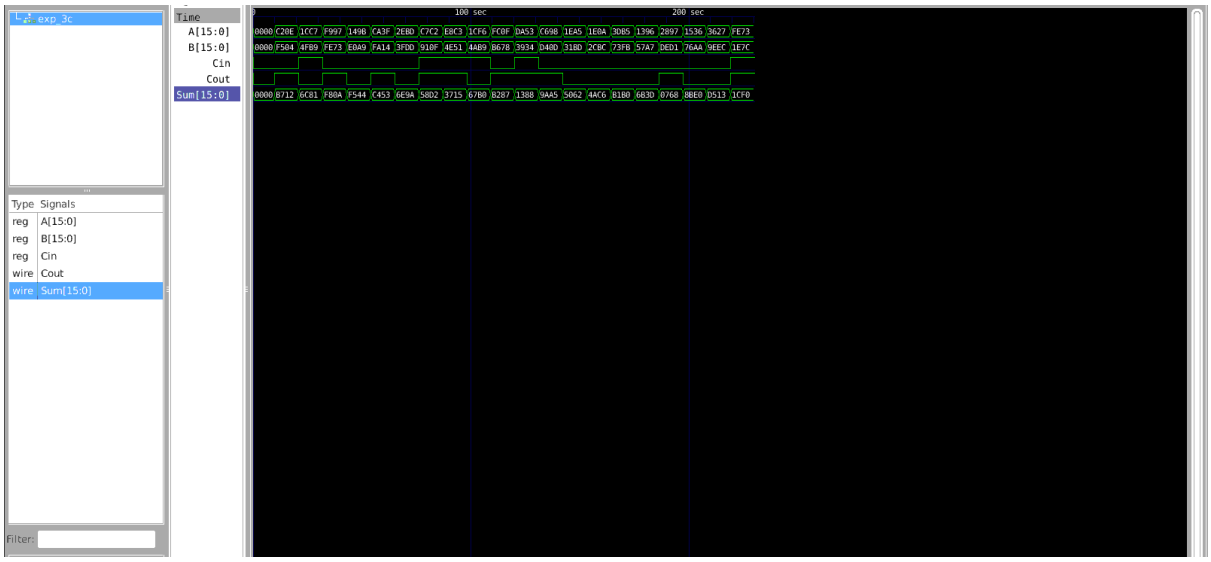
 logic[31:0] seed;
 initial begin
  $dumpfile("dumpfile5.vcd");
  $dumpvars(0,A[15:0],B[15:0],Cin,Sum[15:0],Cout);
  A=0;
  B=0;
  Cin=0;
  seed = 1000;
 end

 integer x;
 always begin
 for(x=1;x<=20;x=x+1)begin
  #10
  {A,B,Cin} = $random(seed) % 8589934592;
  #1
  assert ({Cout,Sum} == (A+B+Cin))
    else $error("Failed: %0d+%0d+%0d != {%0d,%0d}",A,B,Cin,Sum,Cout);
  end
  #10
  $finish;
 end
```

A[15:0] B[15:0] Cin → A[15:0] B[15:0] Cin koggestone Cout S[15:0] → x1 Cout Sum[15:0]

Testbench

GTKWaves



GTKWaves